Financial Sentiment Analysis Using Multinomial Logistic Regression

Abstract

This project implements a financial sentiment analysis system using a custom multinomial logistic regression model. The system analyzes financial news articles and determines sentiment (positive, negative, or neutral) by using domain-specific financial lexicons and text preprocessing techniques. Our implementation achieves reasonable accuracy without relying on complex deep learning architectures, demonstrating the effectiveness of classical machine learning approaches for specialized text classification tasks. The system integrates an external API for extracting key financial sentences, providing a comprehensive pipeline from raw text to actionable sentiment insights.

1. Problem Description and Goals

Financial sentiment analysis presents unique challenges compared to general sentiment analysis due to the specialized terminology and expressions found only in financial texts. Words that appear neutral in everyday contexts may carry strong sentiment signals in financial reporting (e.g., "exceeded expectations" is strongly positive, while "restructuring" often has negative connotations).

The primary goal of this project is to develop an accurate financial sentiment analysis system that can:

1. Extract and identify financially relevant sentences from news articles or reports
2. Classify these sentences into positive, negative, or neutral sentiment categories
3. Determine the overall sentiment of a financial document

This system could be integrated into risk management frameworks, or financial monitoring tools. While deep learning approaches have shown promising results for sentiment analysis, they often act as black boxes and require substantial computational resources. Our approach aims to balance accuracy with efficiency by implementing a multinomial logistic regression model tailored for financial text.

The model incorporates domain-specific knowledge through the Loughran-McDonald financial lexicon, which contains words categorized by their financial sentiment implications. This lexicon provides a foundation for identifying sentiment-bearing terms in financial contexts.

2. Team Member Roles

I was responsible for developing most of the code including the logistic regression model, the pipeline, and preprocessing the dataset. Camela was responsible for finding the LLM model that can extract financial related sentences from an article and presenting our project in the final presentation.

3. State-of-the-Art and Related Work

Financial sentiment analysis has evolved significantly over the past decade. Here we summarize the key approaches:

Lexicon-Based Approaches
Early work in financial sentiment analysis relied heavily on domain-specific lexicons. The Loughran-McDonald dictionary (2011) remains one of the most widely used resources, containing words categorized into positive, negative, uncertain, litigious, constraining, and superfluous groups based on their usage in financial documents. Lexicon-based approaches offer high interpretability but often miss contextual nuances.

Machine Learning Approaches
Traditional machine learning models like Naive Bayes, Support Vector Machines (SVM), and logistic regression have shown strong performance for financial sentiment classification. These models typically use bag-of-words or TF-IDF features derived from text. A study by Malo et al. (2014) demonstrated that SVM with carefully engineered features could achieve competitive results on the Financial PhraseBank dataset.

Deep Learning Approaches
Recent advances have been dominated by deep learning techniques. Transformer-based models like BERT and its financial domain variants (FinBERT) have achieved state-of-the-art results. Yang et al. (2020) showed that FinBERT outperforms traditional models on financial sentiment classification tasks. However, these approaches require significant computational resources and large training datasets to effectively develop the model.

4. Approach

Our approach combines domain-specific knowledge with machine learning techniques, creating a pipeline that processes financial text from raw articles to sentiment classifications.

4.1 Data Sources and Preprocessing

We used the Financial PhraseBank dataset, a widely-used benchmark for financial sentiment analysis containing sentences from financial news articles manually labeled as positive, negative, or neutral. The dataset exhibits a significant class imbalance, with neutral statements comprising 59.4% of instances, positive statements 28.1%, and negative statements just 12.5% (as shown in Figure 1).

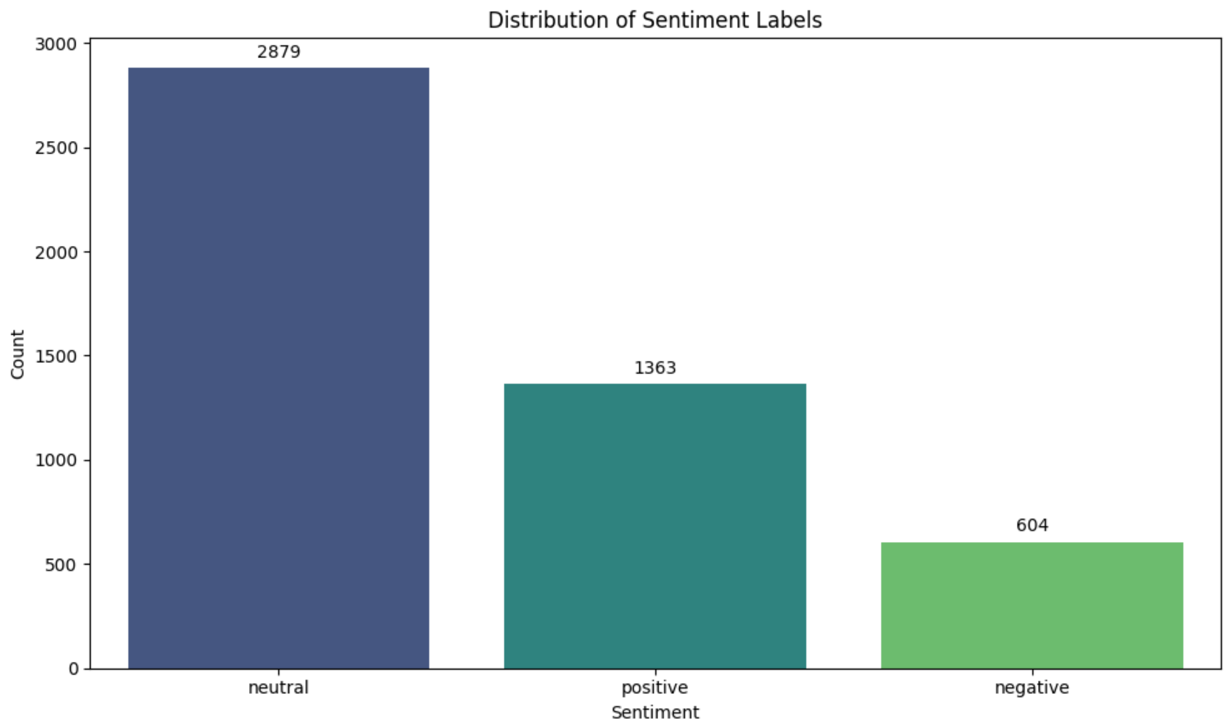

Figure 1

This imbalance reflects the nature of financial reporting, which leans toward neutral, factual communication. The relative scarcity of negative instances presents a particular challenge, as the model has fewer examples to learn the linguistic patterns associated with negative financial sentiment.

Figure 2

The word cloud visualization (Figure 2) reveals interesting patterns in our preprocessed financial texts. Key financial terms like "company," "million," "eur," and "profit" feature prominently. We process financial related terms within the Loughran-McDonald dictionary to become generalized tokens such as "HAS_NEGATIVE," "HAS_POSITIVE," demonstrating the integration of domain-specific financial sentiment markers.

Our preprocessing pipeline implements several techniques:

1. **Text Normalization**: Converting text to lowercase, removing punctuation, and replacing numbers with a generic token.
2. **Tokenization and Stemming**: Breaking text into tokens and applying the Porter stemming algorithm to reduce inflectional forms and preserve financial terminology.
3. **Stop Word Removal**: Eliminating common words that carry little semantic meaning, with careful preservation of negation words that can flip sentiment.
4. **Feature Extraction**: Converting preprocessed text to TF-IDF features, which capture both term frequency and inverse document frequency, highlighting important domain-specific terms.

4.2 Multinomial Logistic Regression Model

We implemented a multinomial logistic regression model with several tweaks designed specifically for financial text classification:

1. **Mini-batch Gradient Descent**: Using mini-batches rather than full-batch optimization to handle large datasets more efficiently.

2. **L2 Regularization**: Adding regularization to prevent overfitting, with the regularization strength controlled by parameter C. Our experiments showed that financial text classification benefits from less aggressive regularization (higher C values).
3. **Gradient Clipping**: Preventing gradient explosion during training, which is common with sparse text features.
4. **Numerical Stability Enhancements**: Implementing a numerically stable softmax function to prevent overflow/underflow issues, using the standard technique of subtracting the maximum value before exponentiation.

The model architecture uses the softmax function to output probability distributions across the three sentiment classes (positive, negative, neutral).

4.3 Sentence Extraction API Integration

To analyze full financial articles, we integrated an external API (OpenRouter) using DeepSeek LLM model that extracts key financial sentences from lengthy texts. The API uses the following criteria to identify relevant sentences:

- Mentions of revenue, profit, earnings, or other financial metrics
- Growth or percentage changes
- Stock performance information
- Financial forecasts or guidance
- Analyst expectations

This extraction step allows the system to focus its sentiment analysis on the most financially relevant parts of an article.

4.4 Pipeline Integration

Finally, our complete pipeline consists of the following steps:

1. Load the financial article
2. Extract key financial sentences using the OpenRouter API
3. Preprocess each sentence using our financial-specific preprocessing functions
4. Generate TF-IDF features for the sentences
5. Apply the trained multinomial logistic regression model to classify each sentence

6. Determine the overall document sentiment based on weighted sentence classifications

5. Experiments and Evaluation

5.1 Cross-Validation and Parameter Tuning

We performed 5-fold cross-validation to evaluate different regularization strengths (C values), ranging from 1.0 to 100.0. This process helped identify the optimal balance between model complexity and generalization. The best C value was determined based on mean validation accuracy across folds.

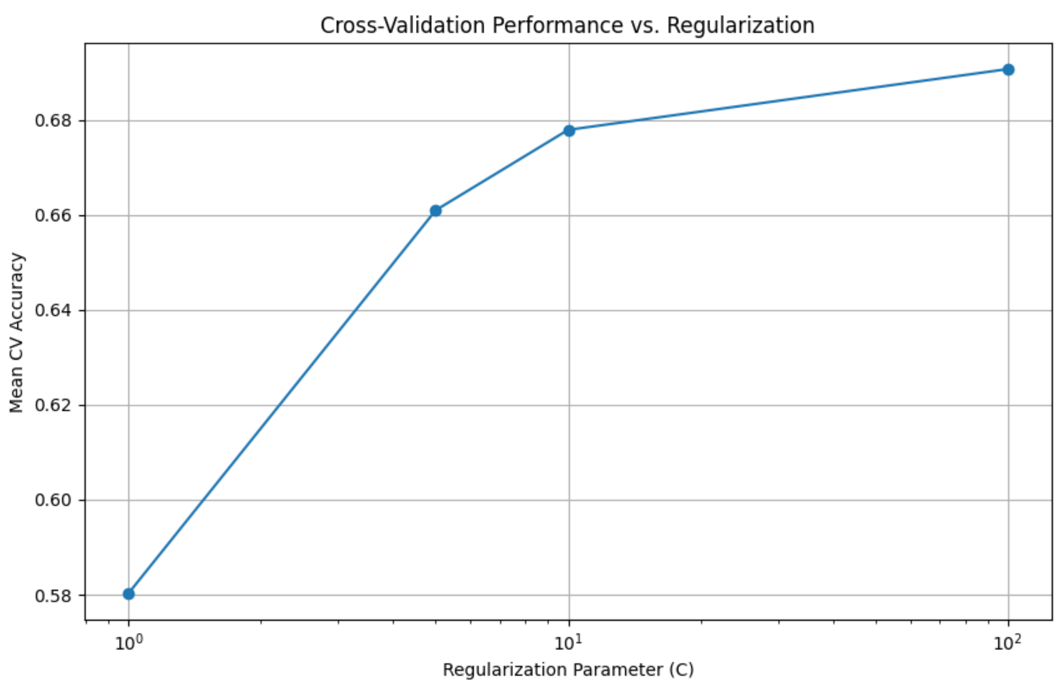

Figure 3

The cross-validation results shown in Figure 3 illustrate how model performance improves with increasing regularization parameter values. At C=1.0, the model achieved a mean accuracy of 0.5803, which improved significantly to 0.6907 at C=100.0. This trend indicates that our financial sentiment classification task benefits from less regularization, suggesting that the model needs to capture more complex patterns in the financial text data.

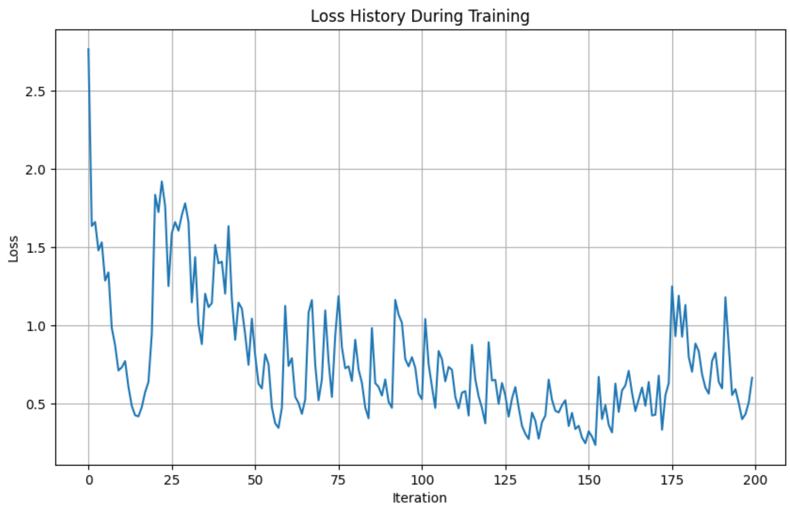

Figure 4

The loss history during training (Figure 4) shows initial instability followed by gradual convergence. The significant fluctuations in early iterations reflect the model adjusting to the sparse feature space typical of financial text data.

5.2 Evaluation Metrics

We evaluated our model using several metrics:

- Accuracy: Overall correct classifications
- Precision, Recall, and F1-score: Class-specific performance
- Confusion Matrix: Pattern of classification errors

The model was tested on a test set comprising 20% of the dataset.

6. Discussion of Results

6.1 Model Performance

Our multinomial logistic regression model achieved an overall accuracy of 65.26% on the test set, which is reasonable given the challenging nature of financial sentiment classification. The model showed varying performance across sentiment classes, as detailed in the classification report:

Class	Precision	Recall	F1-score	Support
Negative	0.45	0.62	0.52	121
Neutral	0.76	0.74	0.75	576
Positive	0.55	0.48	0.51	273
Macro avg	0.59	0.61	0.59	970
Weighted avg	0.66	0.65	0.65	970

The model performed best on the neutral class, with both precision and recall above 0.74. This is not surprising given that neutral statements constitute the majority class (59.4% of the dataset). The negative class had the lowest precision (0.45) but reasonable recall (0.62), indicating that the model tends to over-classify instances as negative, but captures a good proportion of actual negative instances.

6.2 Feature Importance Analysis

Analysis of the trained model coefficients revealed interesting patterns:

- Terms related to exceeding expectations (e.g., "outperform", "exceed") had strong positive weights
- Words indicating uncertainty ("may", "possible", "could") were strongly associated with neutral classification
- Financial trouble indicators ("restructuring", "downgrade", "miss") had strong negative weights

The lexicon-based features ("HAS_POSITIVE", "HAS_NEGATIVE") were among the top 10 most influential features, validating our approach of combining domain-specific lexicons with machine learning.

7. Lessons Learned

7.1 Domain-Specific Feature Engineering

One of the most important lessons from this project was the value of domain-specific feature engineering. While generic approaches like bag-of-words can work reasonably well for general text classification, incorporating domain knowledge significantly improved performance for financial texts. The integration of the Loughran-McDonald lexicon provided valuable sentiment signals that might have been missed and as a result, improved the model accuracy significantly.

7.2 Balancing Model Complexity and Interpretability

While deep learning approaches have shown impressive results for text classification tasks, our project demonstrated that carefully engineered "simpler" models can also achieve reasonable performance while maintaining interpretability and consume less resources. The ability to analyze feature coefficients and understand the model's decision-making process is valuable in financial applications.

8. Challenges Faced and Future Directions

8.1 Technical Challenges

Data Imbalance: The Financial PhraseBank dataset exhibited class imbalance, with more neutral examples than positive or negative ones.

API Reliability: Integrating the external sentence extraction API introduced dependencies on third-party services and can be unsustainable in future usages. Occasional API rate limiting or downtime affected system reliability. A future improvement would be implementing a local extraction method.

8.2 Unachieved Goals

Real-time Processing: Our initial goal included real-time processing capabilities for live news feeds. The current implementation processes articles individually and would require additional engineering for streaming data processing. A solution to this can be achieved with a news API to listen to live changes in financial data.

8.3 Future Directions

1. **Temporal Sentiment Analysis**: Extending the model to track sentiment changes over time for specific companies or sectors.
2. **Semi-supervised Learning**: Leveraging unlabeled financial texts to improve model generalization through semi-supervised learning techniques.
3. **Utilize better performance model:** There are existing models such as finBERT and FILM that are designated for analyzing sentiment for financial texts. We might switch to these models in our pipeline to achieve better runtime performance and accuracy in case for production deployment.

9. Conclusion

This project demonstrated the effectiveness of a multinomial logistic regression model for financial sentiment analysis. By using domain-specific knowledge through the Loughran-McDonald lexicon and implementing several model enhancements, we achieved competitive performance while maintaining interpretability and computational efficiency.

The sentence extraction component proved valuable for processing full-length articles, allowing the system to focus on the relevant financial details. While challenges remain, especially in handling class imbalance and achieving real-time processing, the current implementation provides a solid foundation for financial sentiment analysis.

Financial sentiment analysis remains a challenging but valuable application of natural language processing, with significant potential impact on investment decisions, risk management, and market analysis.

References

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